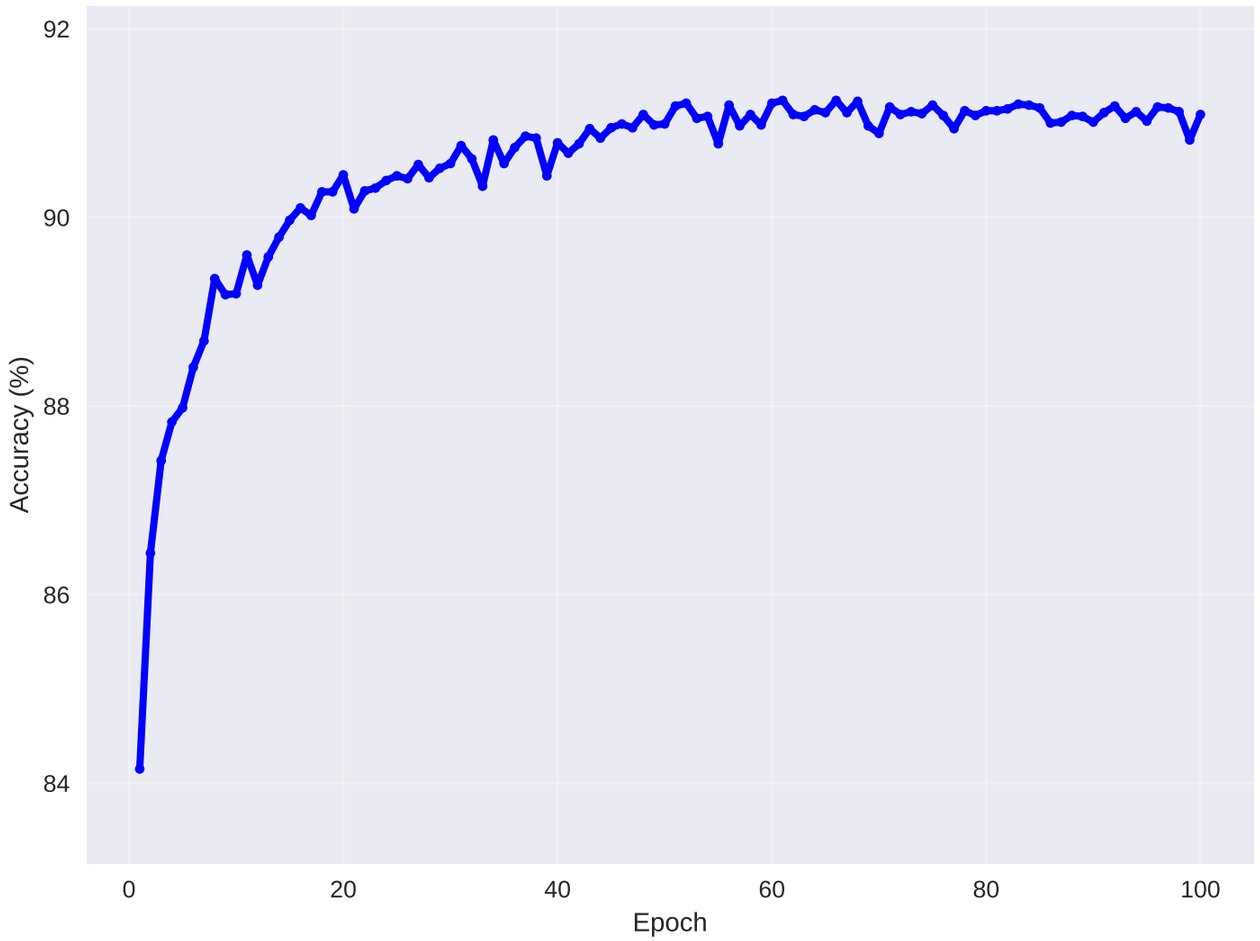
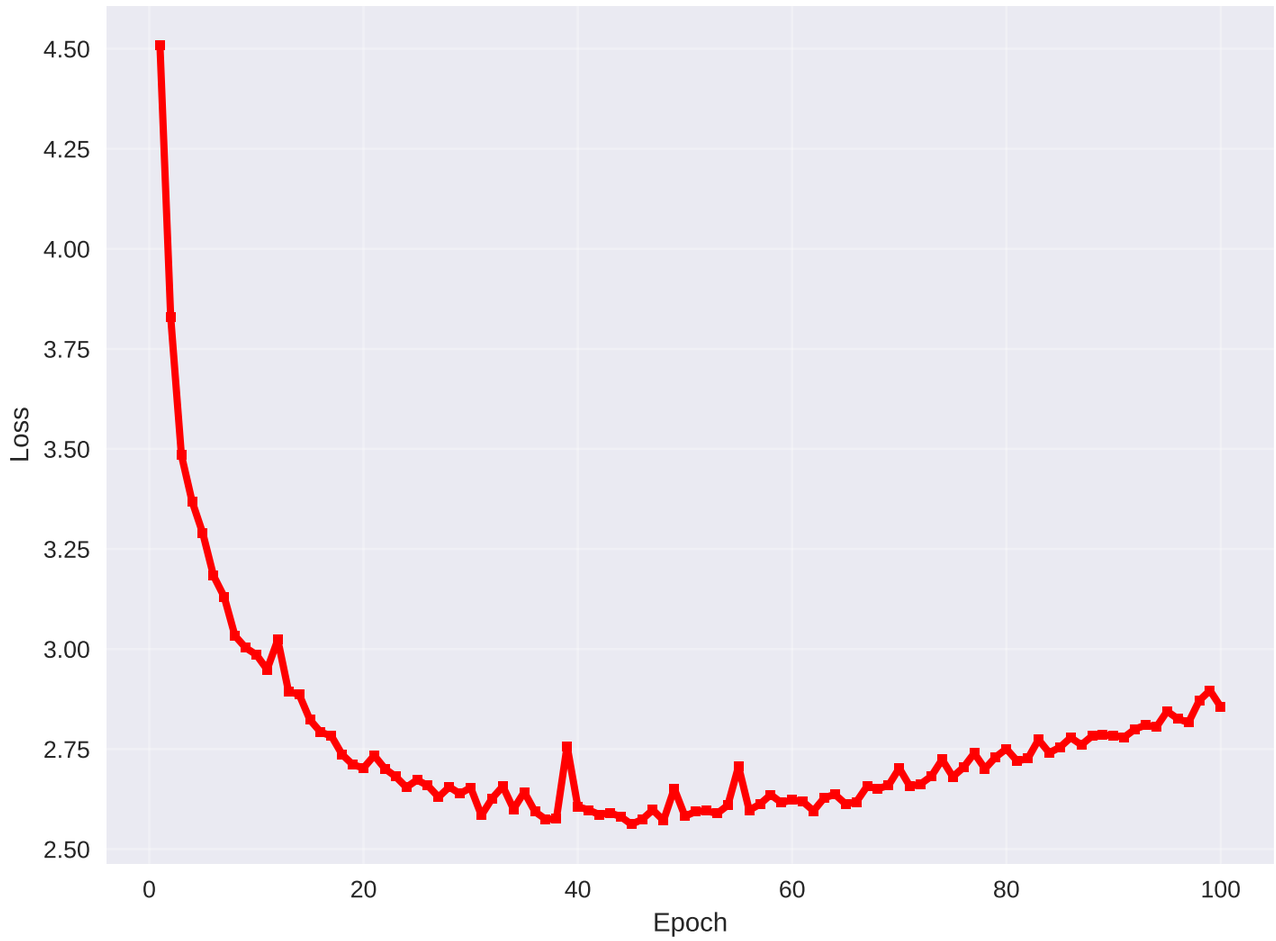
Federated Learning - Main Metrics Overview

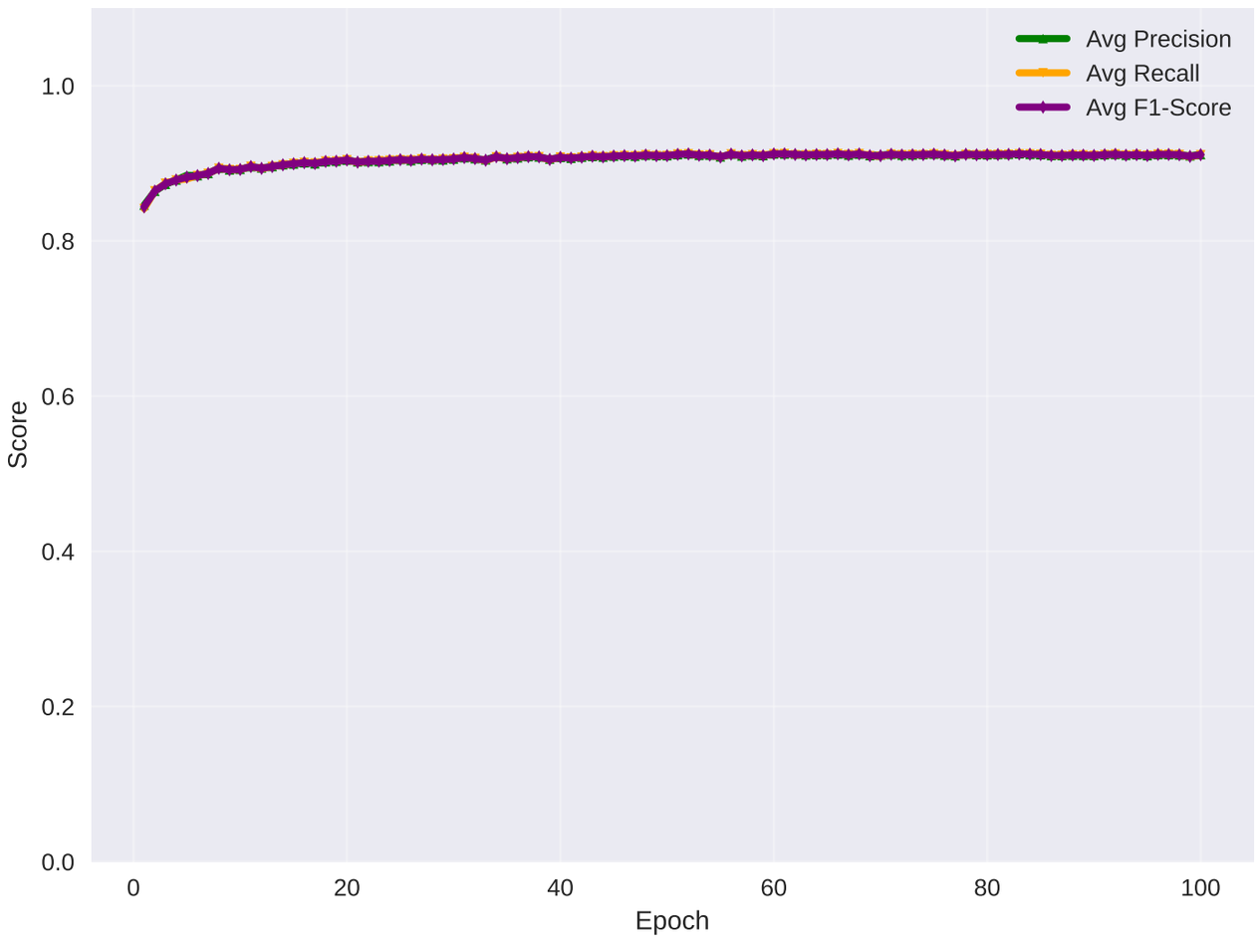
□ Global Model Accuracy Evolution



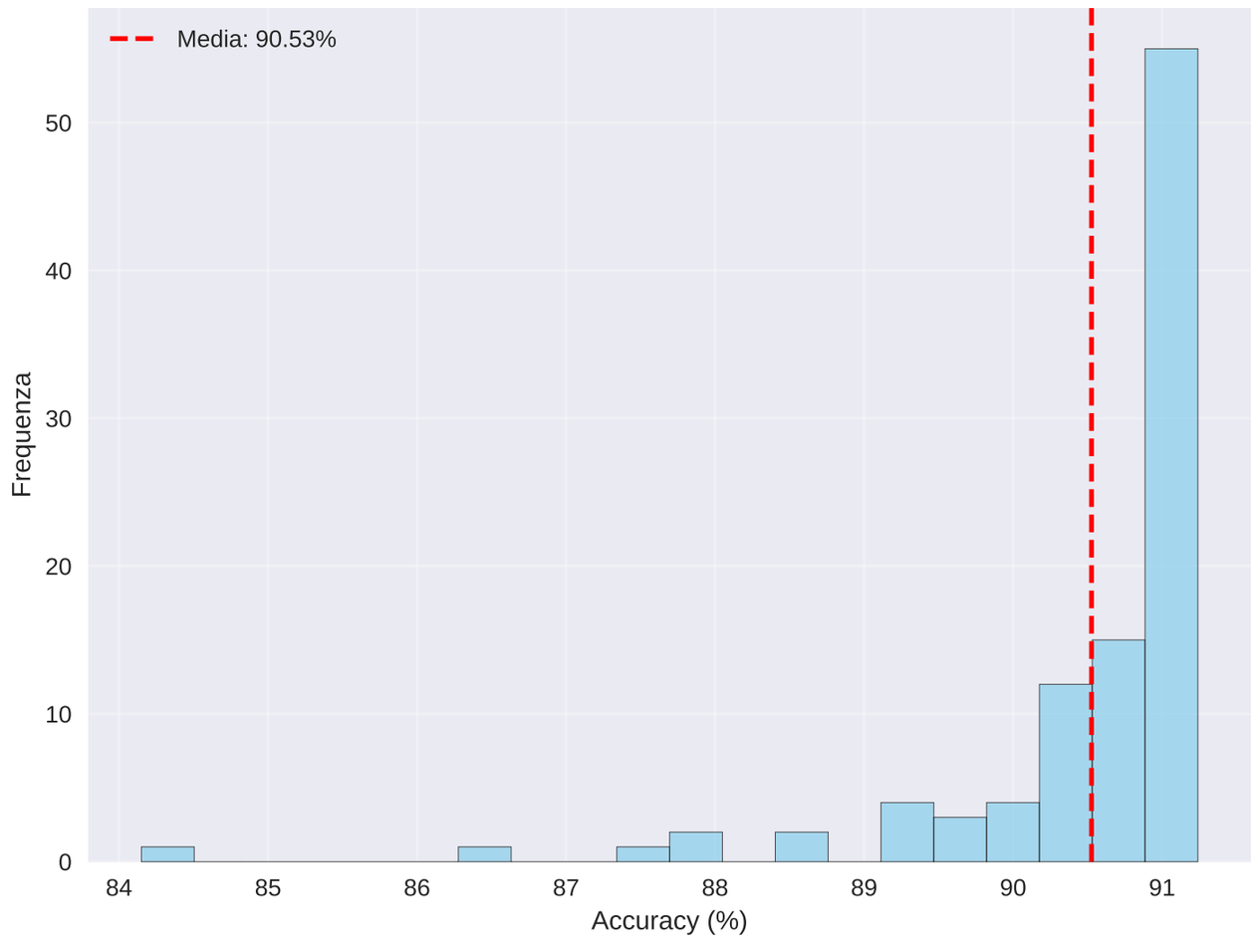
□ Global Model Loss Evolution



□ Average Precision, Recall & F1-Score

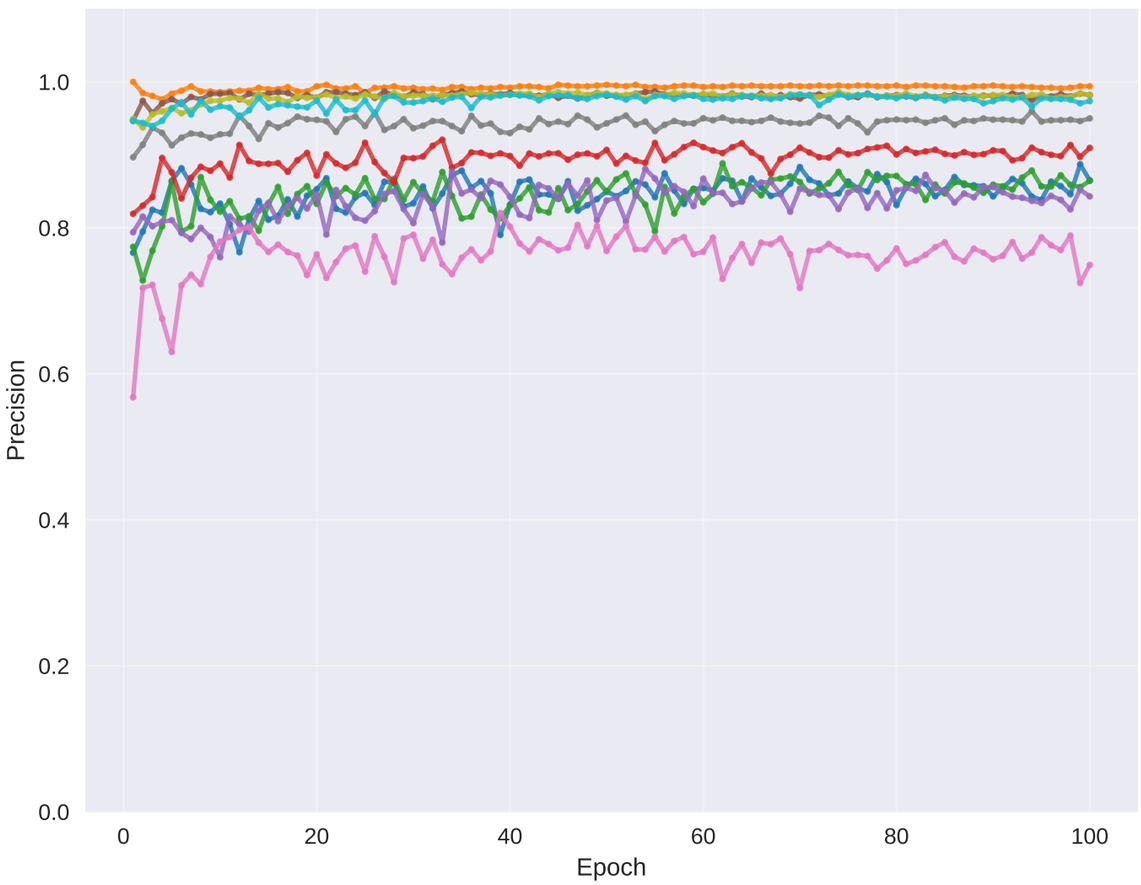


□ Distribuzione Accuracy

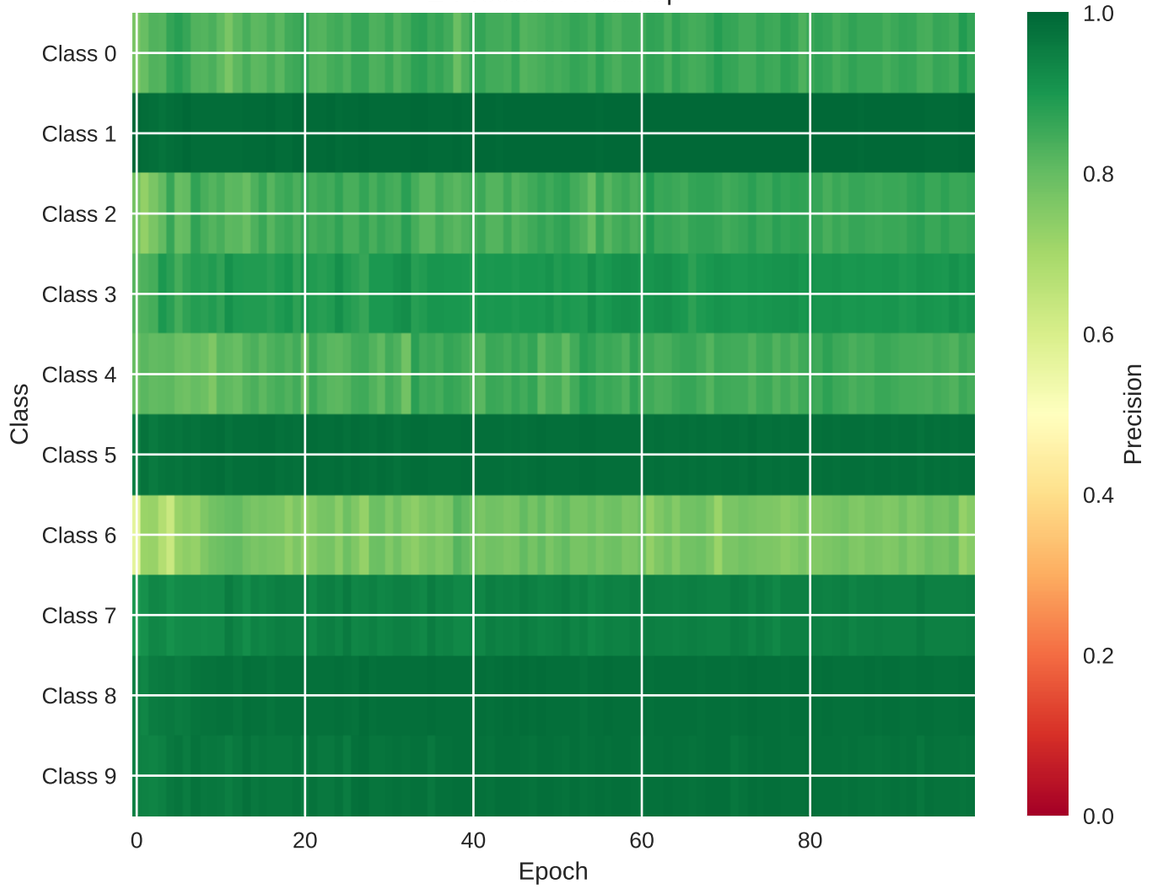


Per-Class Precision Analysis

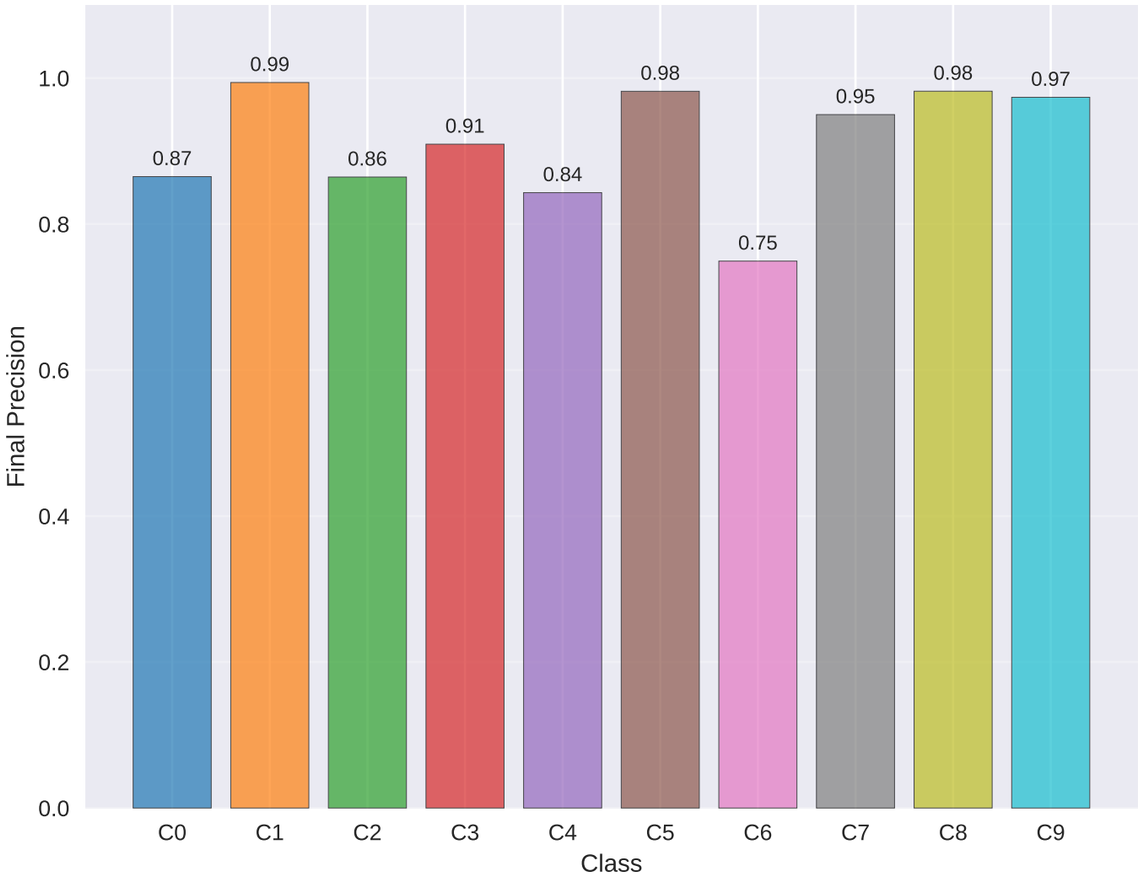
□ Per-Class Precision Evolution



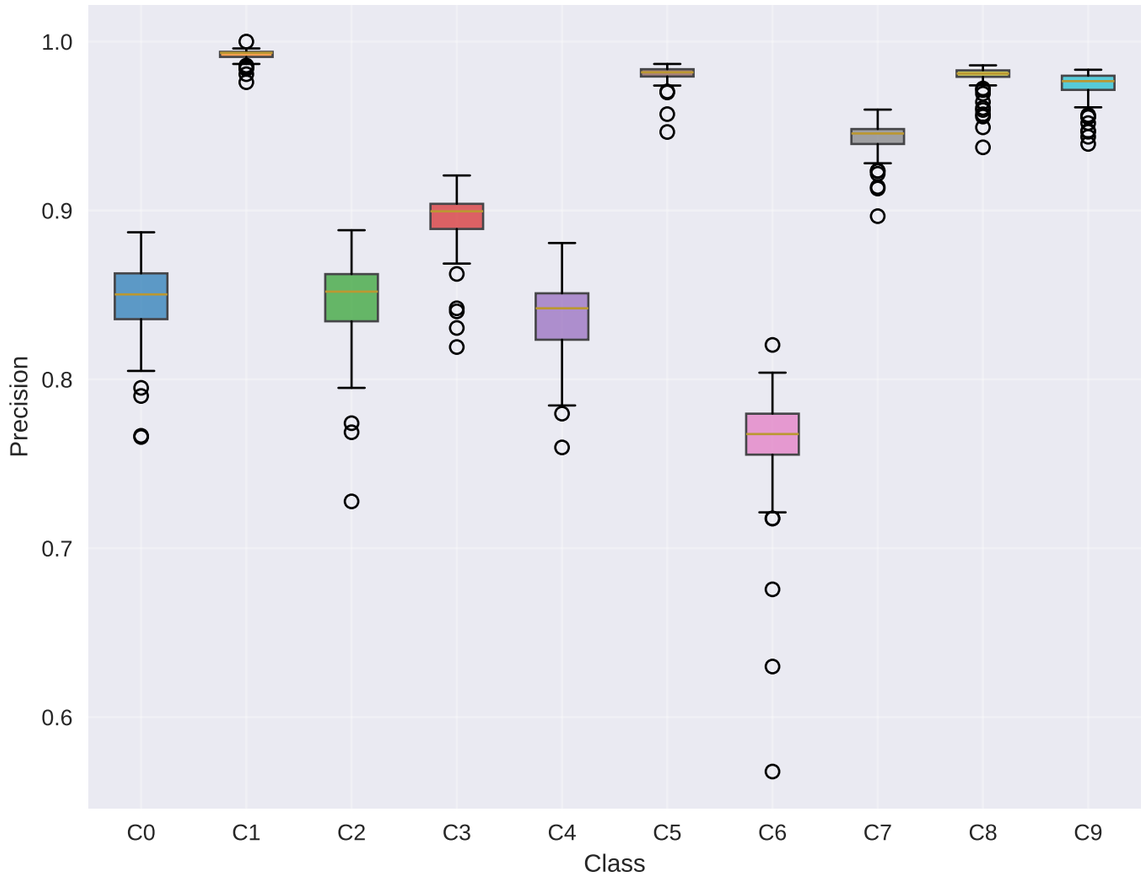
□ Precision Heatmap



□ Final Precision per Class

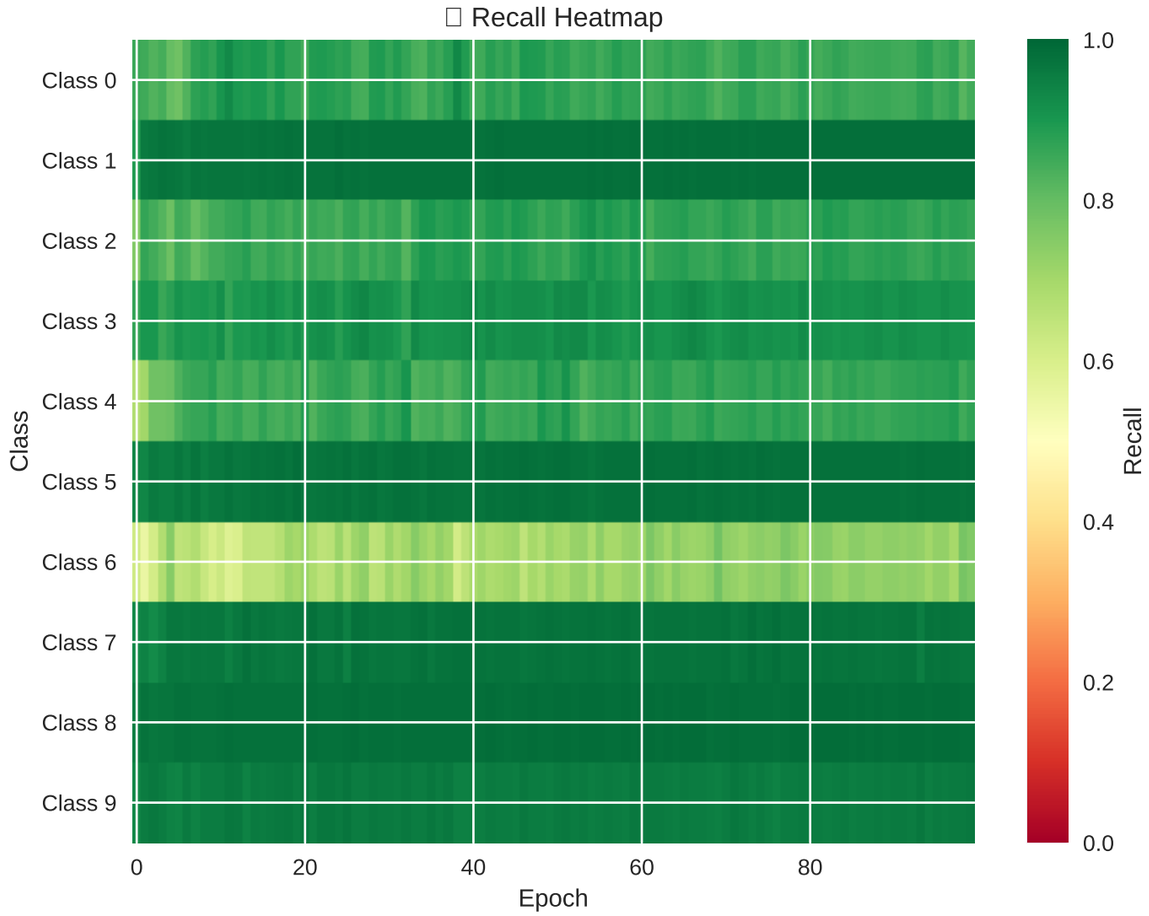


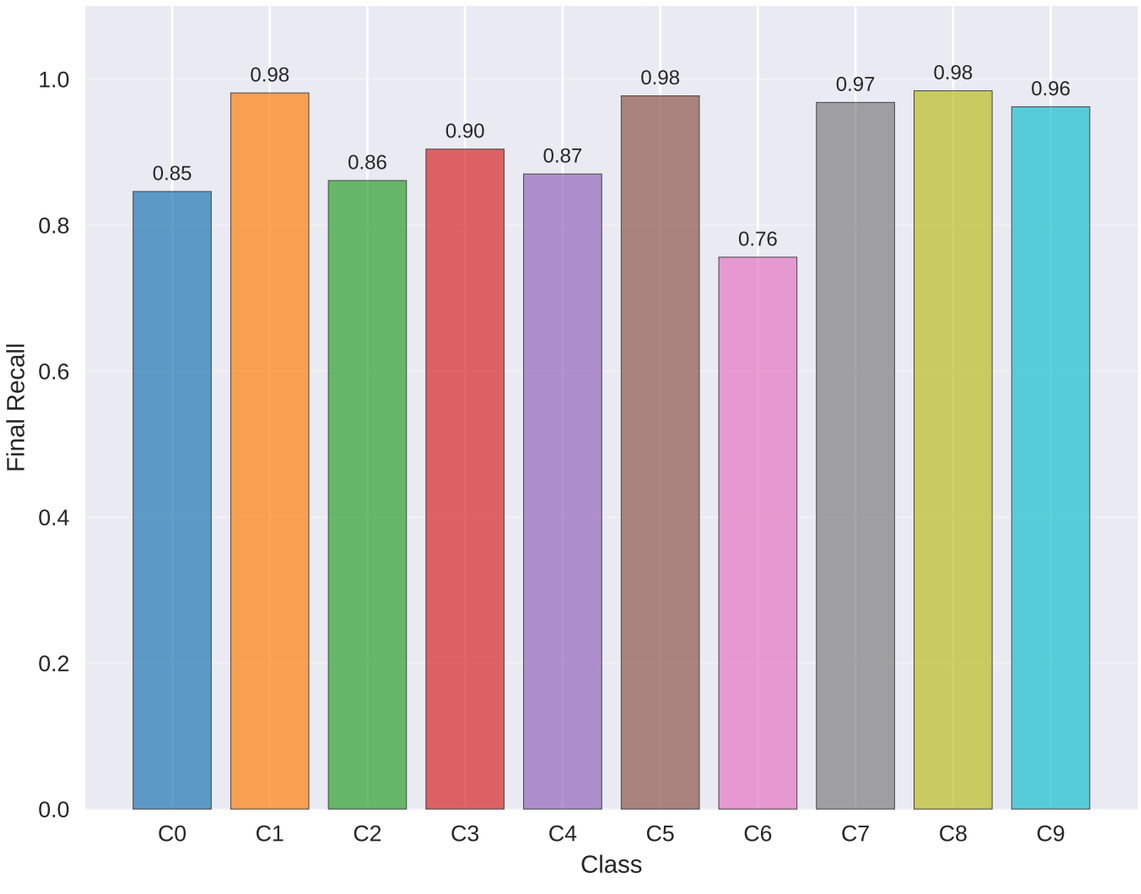
□ Precision Distribution per Class

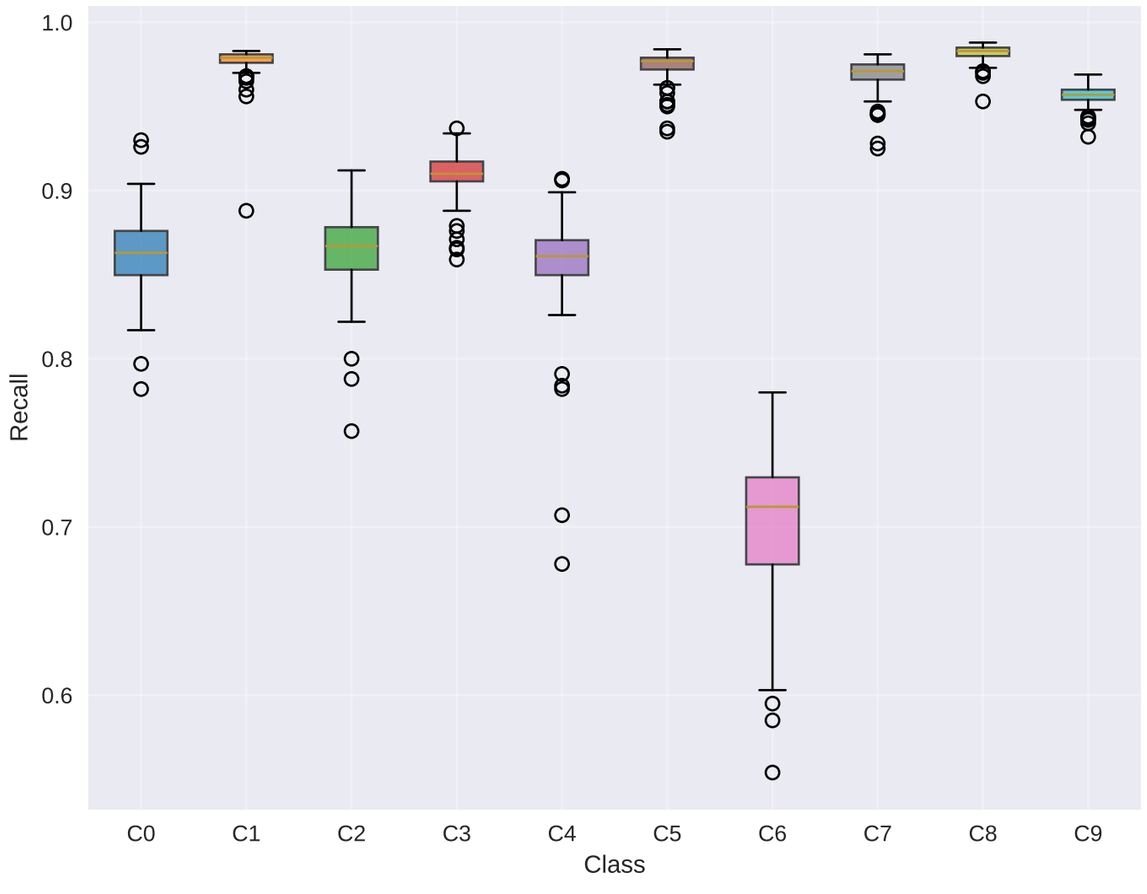


Per-Class Recall Analysis



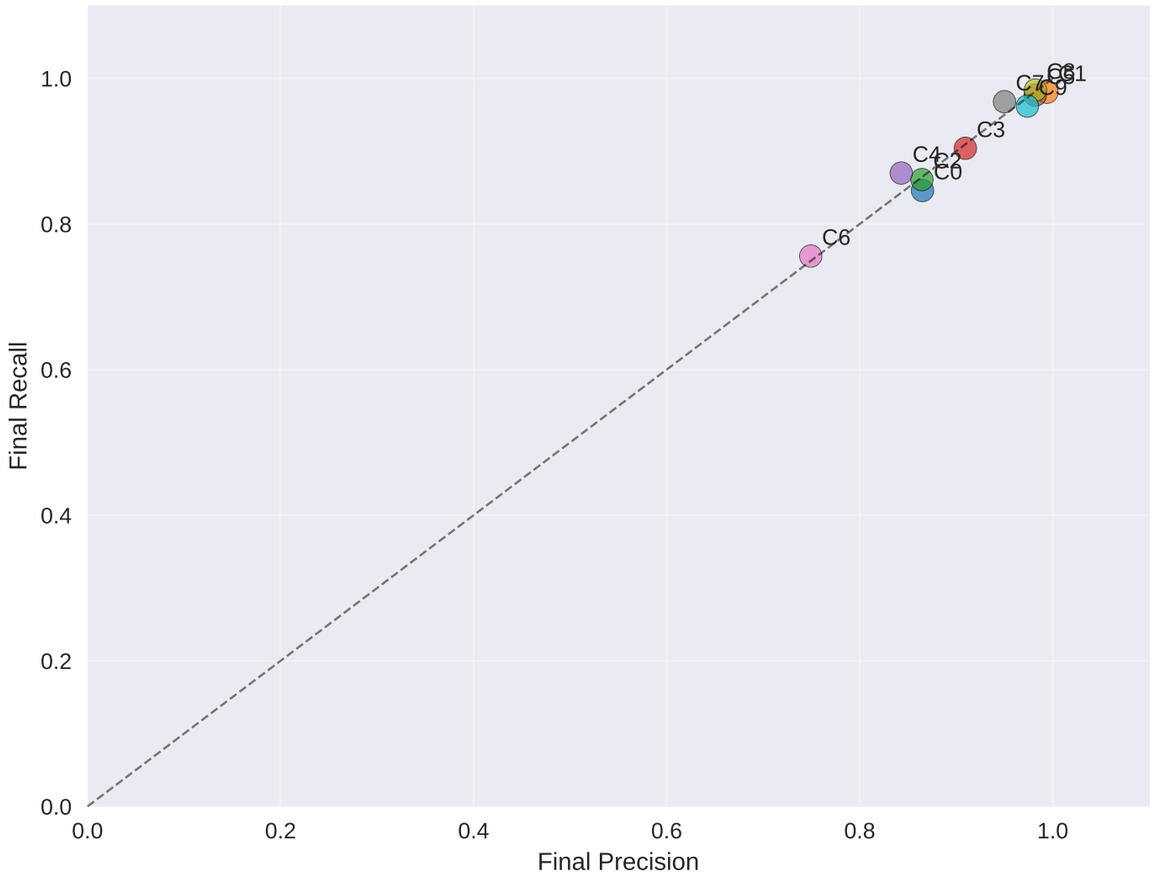




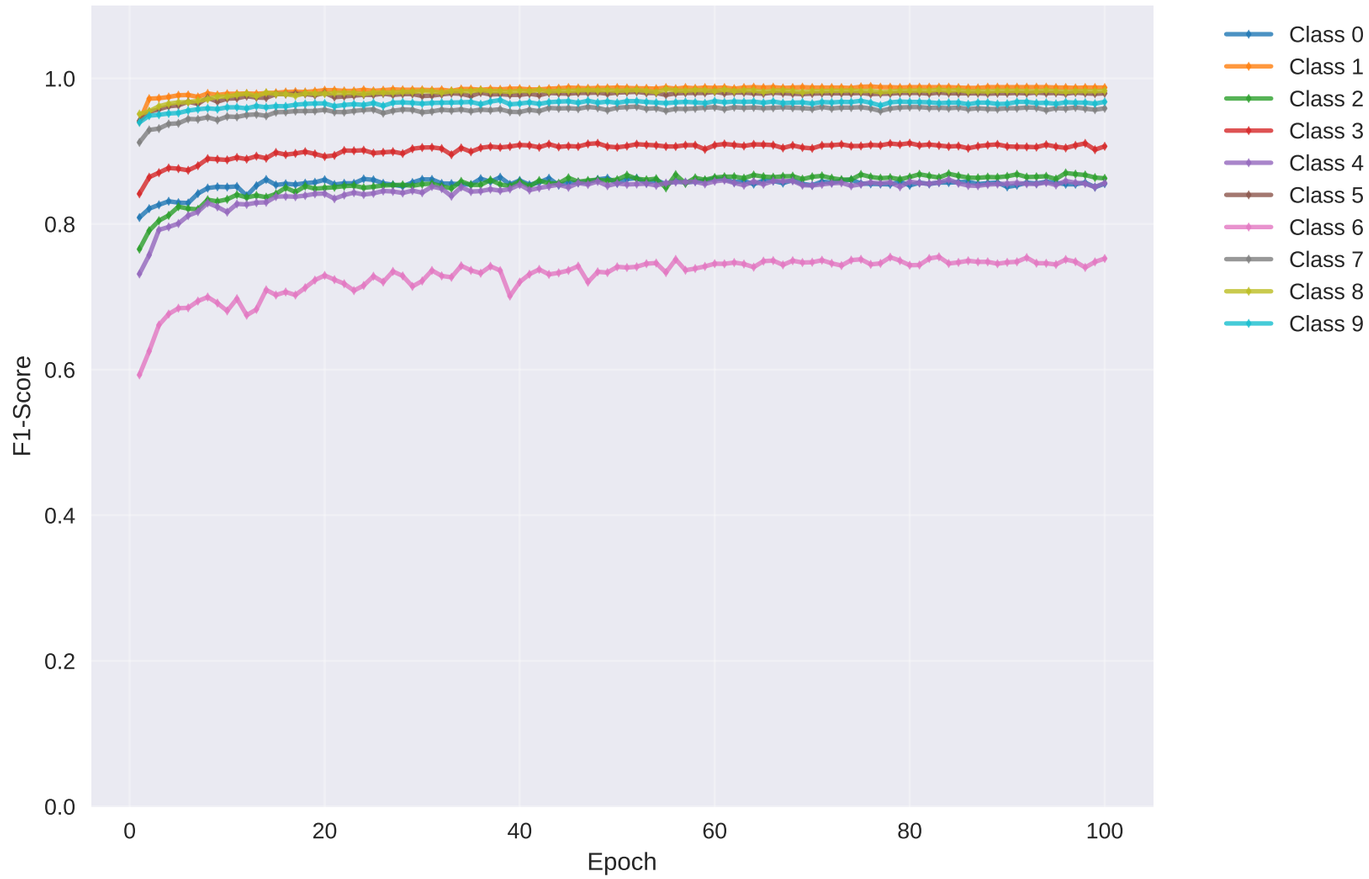


Comparative Analysis

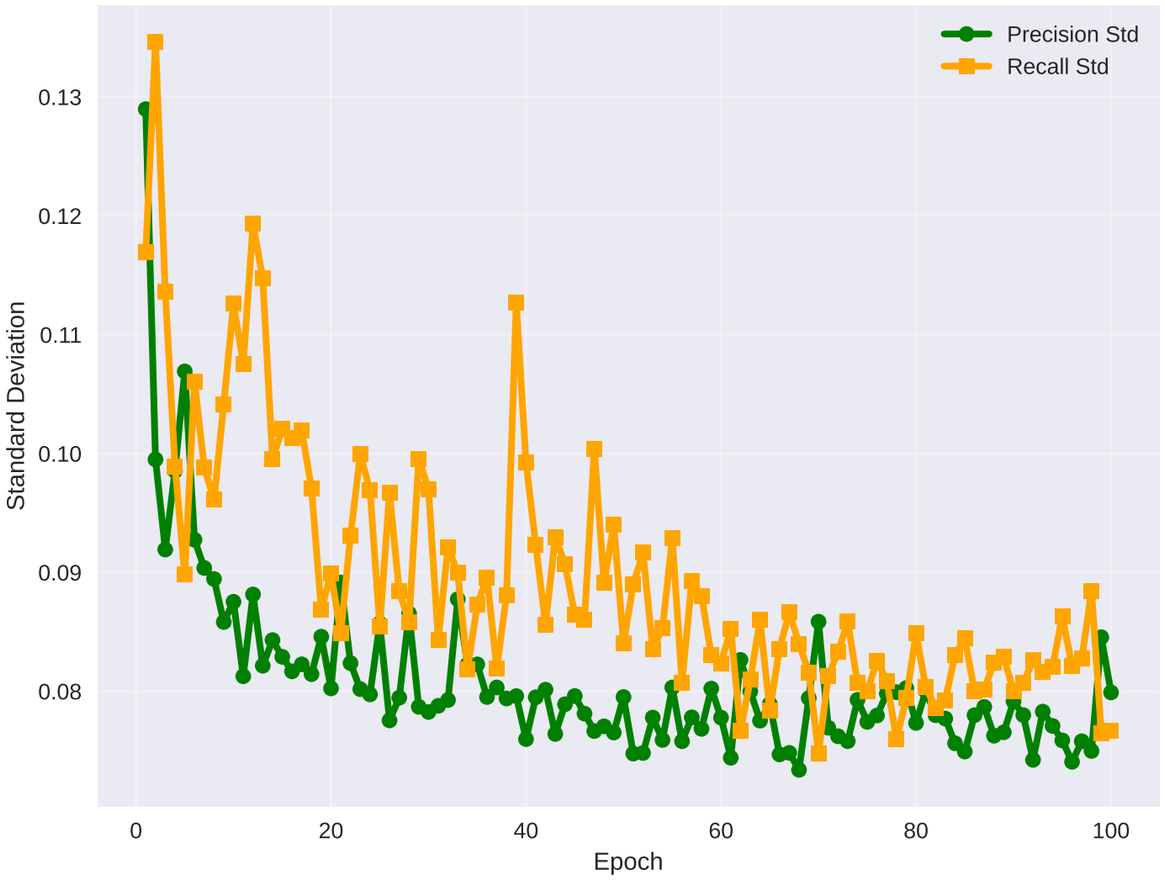
□ Precision vs Recall (Final Values)



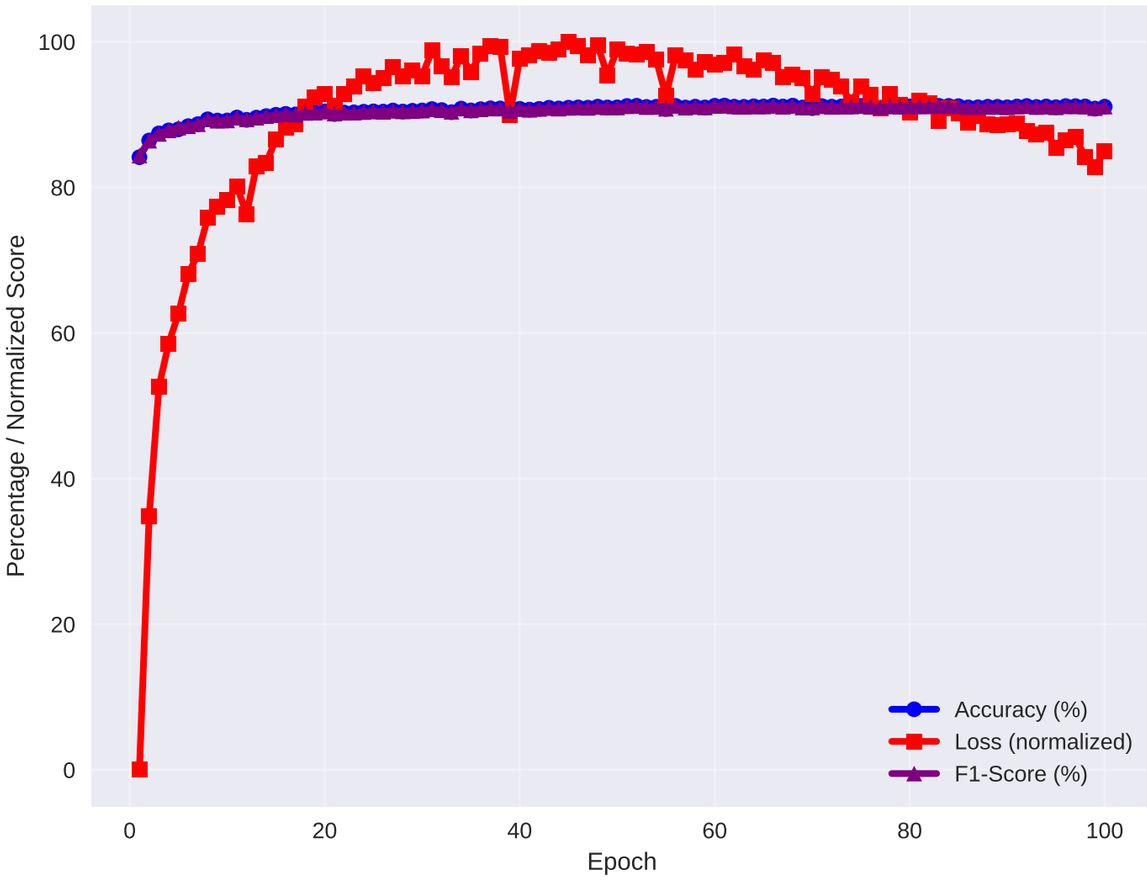
□ Per-Class F1-Score Evolution



□ Inter-Class Variance Evolution



□ Learning Curves Comparison



Performance Summary Report

```
FEDERATED LEARNING - PERFORMANCE SUMMARY
========================================

FILE ANALIZZATO: 3000_results.csv
DATA ANALISI: 2025-08-31 02:30:20
EPOCHE TOTALI: 100

METRICHE FINALI:
• Accuracy: 91.09%
• Loss: 2.8557
• Average Precision: 0.911
• Average Recall: 0.911
• Average F1-Score: 0.911

EVOLUZIONE TRAINING:
• Miglioramento Accuracy: +6.94%
• Riduzione Loss: +1.6535
• Stabilità Accuracy (CV): 0.0120

PERFORMANCE PER CLASSE:
• Migliore Precision: Classe 1 (0.994)
• Peggiore Precision: Classe 6 (0.749)
• Migliore Recall: Classe 8 (0.984)
• Peggiore Recall: Classe 6 (0.756)

ANALISI DETTAGLIATA PER CLASSE:
• Classe 0: P=0.865, R=0.846, F1=0.855
• Classe 1: P=0.994, R=0.981, F1=0.987
• Classe 2: P=0.864, R=0.861, F1=0.863
• Classe 3: P=0.909, R=0.904, F1=0.907
• Classe 4: P=0.843, R=0.870, F1=0.856
• Classe 5: P=0.982, R=0.977, F1=0.979
• Classe 6: P=0.749, R=0.756, F1=0.753
• Classe 7: P=0.950, R=0.968, F1=0.959
• Classe 8: P=0.982, R=0.984, F1=0.983
• Classe 9: P=0.974, R=0.962, F1=0.968

VALUTAZIONE QUALITATIVA:
• Performance Globale: BUONA
• Stabilità Training: STABILE
• Balance Classes: SBILANCIATO

RACCOMANDAZIONI:
```